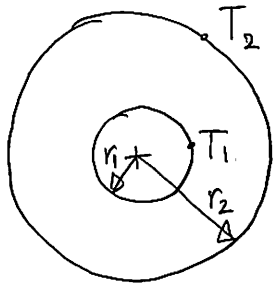


③ Spherical System



Using our spherical heat equation

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = 0$$

$$\int \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = \int 0$$

$$r^2 \frac{\partial T}{\partial r} = C_1 \Rightarrow \int \frac{\partial T}{\partial r} = \int \frac{C_1}{r^2}$$

$$T = -\frac{C_1}{r} + C_2$$

Our boundary conditions are:

$$T_1 = -\frac{C_1}{r_1} + C_2 \quad (1)$$

$$T_2 = -\frac{C_1}{r_2} + C_2 \quad (2)$$

① - ②

$$C_1 = \frac{T_1 - T_2}{\left(\frac{1}{r_2} - \frac{1}{r_1}\right)}$$

$$C_2 = T_2 + \frac{T_1 - T_2}{\left(\frac{1}{r_2} - \frac{1}{r_1}\right)} \cdot \frac{1}{r_2}$$

$$T = \frac{T_1 - T_2}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)} \cdot \frac{1}{r} + \frac{T_1 - T_2}{\left(\frac{1}{r_2} - \frac{1}{r_1}\right)} \cdot \frac{1}{r_2} + T_2$$

$$\boxed{\frac{T - T_2}{T_1 - T_2} = \frac{\left(\frac{1}{r} - \frac{1}{r_2}\right)}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}} \Rightarrow \text{Spherical temperature profile} \quad (26)$$

Calculating our thermal resistance:

$$Q = -k A \frac{\partial T}{\partial r} \quad ; \quad \frac{\partial T}{\partial r} = \frac{T_1 - T_2}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)} \left(-\frac{1}{r^2}\right)$$

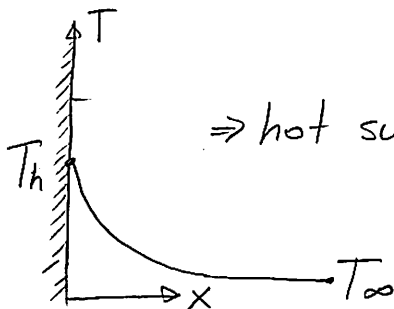
$$= -k (4\pi r^2) \frac{T_1 - T_2}{\frac{1}{r_1} - \frac{1}{r_2}} \left(-\frac{1}{r^2}\right)$$

$$Q = \frac{T_1 - T_2}{\left\{ \frac{\frac{1}{r_1} - \frac{1}{r_2}}{4\pi k} \right\}} = \frac{\Delta T}{R_{sph}}$$

$$\boxed{R_{sph} = \frac{\left| \frac{1}{r_1} - \frac{1}{r_2} \right|}{4\pi k}} \Rightarrow \text{Spherical thermal resistance.}$$

We can do the same thermal resistance analogy for convection heat transfer:

$$Q = h A \Delta T \quad ; \quad h = \text{heat transfer coefficient [W/m}^2 \cdot \text{K]} \\ A = \text{area}$$



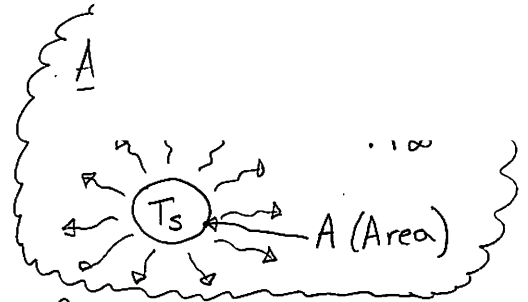
$\Rightarrow$ hot surface exchanging heat with a fluid

Newton developed this concept in the 1700's. Known as Newton's law of cooling.

$$Q = h A \Delta T = \frac{\Delta T}{(1/hA)} \Rightarrow \boxed{R_{Th, conv} = \frac{1}{hA}}$$

$\hookrightarrow$ Thermal resistance associated with convection

Radiation thermal resistance:



$$Q_{\text{rad}} = \underbrace{\epsilon}_{\text{Emissivity}} \underbrace{\sigma}_{\text{Stefan-Boltzmann Constant } (5.67 \times 10^{-8})} A (T_s^4 - T_\infty^4) \Rightarrow \text{Stefan Boltzmann Law (Black Body)}$$

Factoring: $(T_s^2 + T_\infty^2)(T_s + T_\infty)(T_s - T_\infty)$

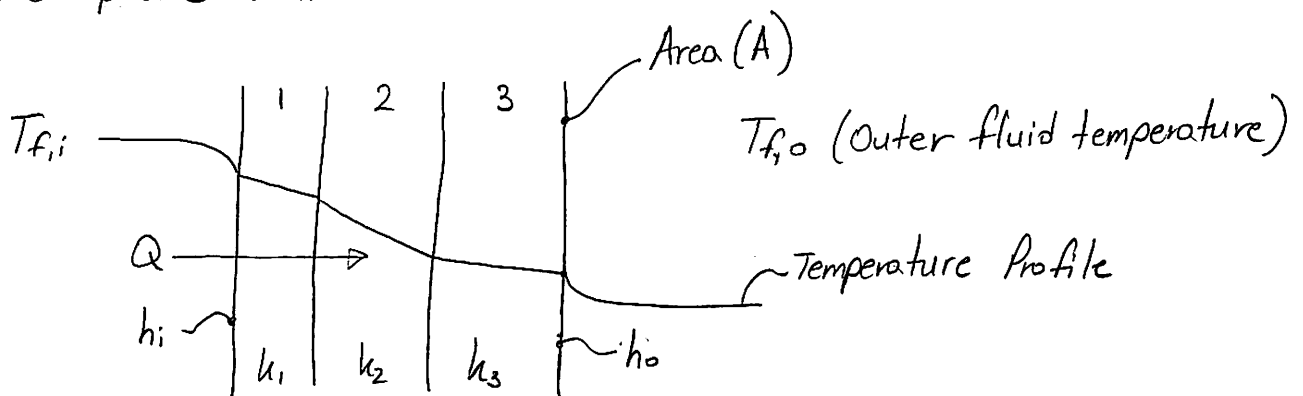
$$Q_{\text{rad}} = \underbrace{\epsilon \sigma A (T_s^2 + T_\infty^2)(T_s + T_\infty)}_{h_{\text{rad}} \cdot A} \underbrace{(T_s - T_\infty)}_{\Delta T}$$

$$h_{\text{rad}} = \epsilon \sigma (T_s^2 + T_\infty^2)(T_s + T_\infty) \Rightarrow \text{Radiation heat transfer coefficient.}$$

$$R_{\text{Radiation}} = \frac{1}{h_{\text{rad}} A} \Rightarrow \text{Thermal resistance associated with radiation. (nonlinear)}$$

Composite problems

① Composite wall:

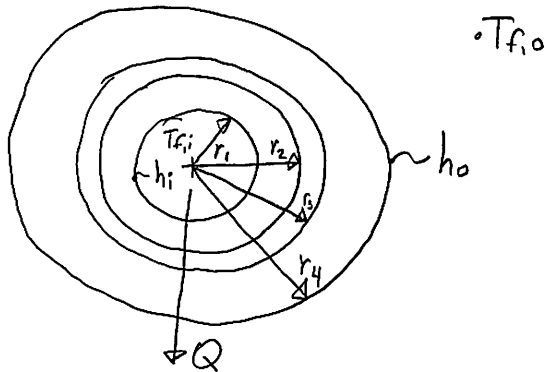


$$Q = \frac{A (T_{f,i} - T_{f,o})}{\frac{1}{h_i} + \frac{1}{h_o} + \sum_{j=1}^n \frac{L_j}{k_j}}$$

L_j = thickness of the j 'th wall
 k_j = thermal conductivity of the j 'th wall

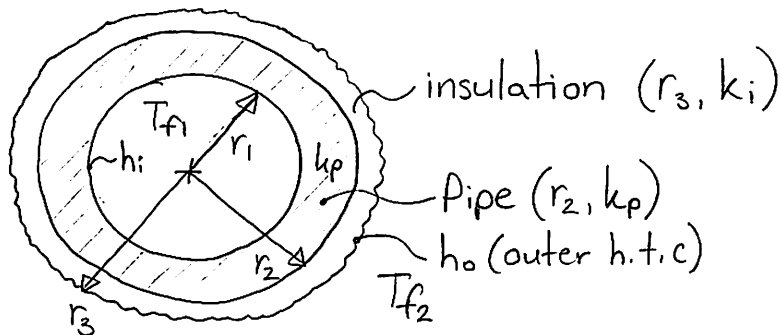
h_i = inner wall heat transfer coeff.
 h_o = outer wall h.t.c.

② Composite cylinder



$$Q = \frac{2\pi L (T_{f,i} - T_{f,o})}{\frac{1}{h_i r_1} + \frac{1}{h_o r_{n+1}} + \sum_{j=1}^n \frac{\ln(r_{j+1}/r_j)}{k_j}}$$

Critical Thickness of Insulation:



k_i = insulation thermal c.
 k_p = pipe thermal cond.
 $T_{f,i}$ = inner fluid temp.
 $T_{f,o}$ = outer fluid temp.
 h_o = outer h.t.c.
 $r_1 < r_2 < r_3$

Note, this is an optimization problem since by adding insulation:

1) The outer area, for which h_o convects heat away increases $\Rightarrow Q_{loss} \uparrow$

2) The resistance of the insulation increases $\Rightarrow Q_{loss} \downarrow$

$$R_{TOT} = \underbrace{\frac{1}{2\pi r_1 h_i}}_{\text{inner convection}} + \underbrace{\frac{\ln(r_2/r_1)}{2\pi k_p L}}_{\text{pipe cond.}} + \underbrace{\frac{\ln(r_3/r_2)}{2\pi k_i L}}_{\text{insulation cond.}} + \underbrace{\frac{1}{2\pi r_3 h_o L}}_{\text{outer convection}}$$

We need to maximize R_{TOT} with respect to r_3

$$\frac{\partial R_{TOT}}{\partial r_3} = 0$$

$$\frac{1}{2\pi k_i L} \frac{\partial}{\partial r_3} \ln(r_3/r_2) + \frac{1}{2\pi h_o L} \frac{\partial}{\partial r_3} \left(\frac{1}{r_3} \right) = 0$$

$$\frac{1}{k_i} \frac{\partial}{\partial r_3} \ln\left(\frac{r_3}{r_2}\right) + \frac{1}{h_o} \frac{\partial}{\partial r_3} \left(\frac{1}{r_3}\right) = 0$$

$$\frac{1}{k_i} \cdot \frac{k_i}{r_3^2} + \frac{1}{h_o} \left(-\frac{1}{r_3^2}\right) = 0$$

$$\frac{1}{k_i} - \frac{1}{h_o r_3} = 0$$

$$\boxed{r_{3,crit} = \frac{k_i}{h_o}} \Rightarrow \text{Critical thickness of insulation for a pipe.}$$

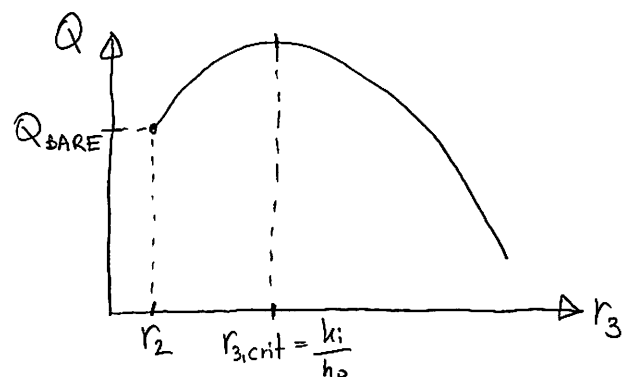
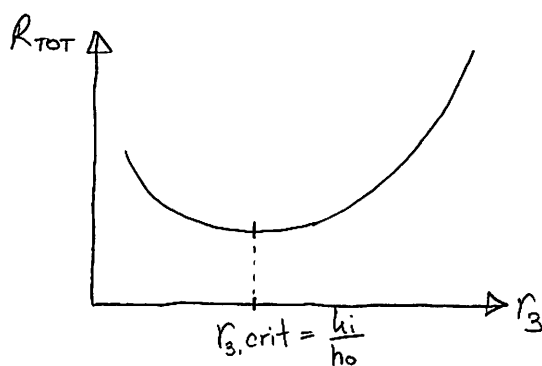
Now we need to check to make sure this is a minimum for Q_{TOT}

$$\begin{aligned} \frac{\partial^2 R_{TOT}}{\partial r_3^2} &= \left[\frac{\partial}{\partial r} \left(\frac{1}{k_i r_3} \right) - \frac{\partial}{\partial r} \left(\frac{1}{h_o r_3^2} \right) \right] \frac{1}{2\pi L} \\ &= \left(-\frac{1}{k_i r_3^2} + \frac{2}{h_o r_3^3} \right) \frac{1}{2\pi L} \end{aligned}$$

Substitute in $r_{3,crit} = \frac{k_i}{h_o}$ into equation ①

$$\begin{aligned} &= -\frac{1}{k_i \left(\frac{k_i^2}{h_o^2} \right)} + \frac{2}{h_o \left(\frac{k_i^3}{h_o^3} \right)} = \left(-\frac{h_o^2}{k_i^3} + \frac{2h_o^2}{k_i^3} \right) \cdot \frac{1}{2\pi L} \\ &= +\frac{h_o^2}{2\pi k_i^3} > 0 \quad (\text{Always}) \end{aligned}$$

So our $R_{TOT}(r_{3,crit})$ is a global minimum.
No optimum thickness exists, only a critical insulation thickness.



If we do a sample calculation

$$\left. \begin{array}{l} k_i = 0.1 \text{ W/m}\cdot\text{K} \\ h_o = 5 \text{ W/m}^2\cdot\text{K} \end{array} \right\} r_{3, \text{crit}} = \frac{0.1 \text{ W/m}\cdot\text{K}}{5 \text{ W/m}^2\cdot\text{K}} = 0.02 \text{ m} = 2 \text{ cm}$$

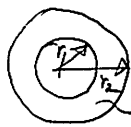
this is a radius!

Note, here $r_{3, \text{crit}} < r_2$ meaning it's OK to insulate.

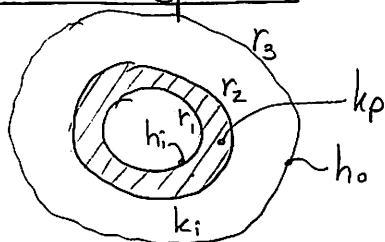
In general : $k_i \sim 0.1 \text{ W/m}\cdot\text{K}$ (common insulating materials)
 $h_{o, \text{minimum}} \sim 5 \text{ W/m}^2\cdot\text{K}$ (natural convection)

$r_{\text{crit}} \sim 1 \text{ cm} \Rightarrow$ We usually operate well above this.

∴ We can insulate hot water and steam pipes without worrying about increasing external heat transfer losses.

How about electrical wires? $\Rightarrow$  $\Rightarrow r_1 \sim 1 \text{ mm}$
 Good to insulate to increase losses & cool the wire.

For a sphere:



$$r_1 < r_2 < r_3$$

$$R_{\text{TOT}} = \frac{1}{4\pi r_1^2 h_i} + \frac{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}{4\pi k_p} + \frac{\left(\frac{1}{r_2} - \frac{1}{r_3}\right)}{4\pi k_i} + \frac{1}{4\pi r_3^2 h_o}$$

$$\frac{2R_{\text{TOT}}}{2r_3} = \frac{1}{4\pi k_i} \left(\frac{1}{r_3^2}\right) + \frac{1}{4\pi h_o} \left(-\frac{2}{r_3^3}\right) = 0$$

$$\frac{1}{r_{3, \text{cr}}^2 k_i} - \frac{2}{h_o r_{3, \text{cr}}^3} = 0 \Rightarrow$$

$$r_{3, \text{cr}} = \frac{2k_i}{h_o}$$

↓ Critical insulation thickness for a sphere.

Heat Generation (Slab)

Steady State, 1D, constant properties, Uniform $\dot{Q}'''$

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{Q}''' = \rho C_p \frac{\partial T}{\partial t} \quad \text{S.S.}$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\dot{Q}'''}{k} = 0 \quad (1)$$

$$T(x=L) = T_w$$

$$\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0$$

Integrating (1)

$$\frac{\partial T}{\partial x} = -\frac{\dot{Q}'''}{k}x + C_1$$

$$\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0 = -\frac{\dot{Q}'''}{k}(0) + C_1 = 0 \Rightarrow C_1 = 0$$

$$T(x) = -\frac{\dot{Q}'''x^2}{2k} + C_2$$

$$T(x=L) = T_w = -\frac{\dot{Q}'''L^2}{2k} + C_2 \Rightarrow C_2 = T_w + \frac{\dot{Q}'''L^2}{2k}$$

$$T(x) = T_w + \frac{\dot{Q}'''L^2}{2k} \left(1 - \left(\frac{x}{L} \right)^2 \right)$$

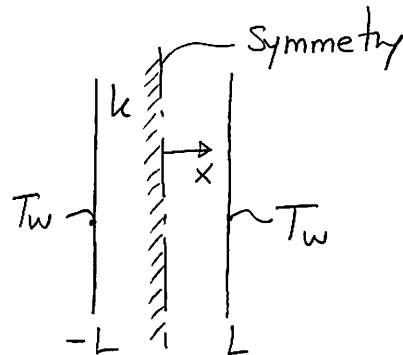
Where is $T_{\max}$?

$$\frac{\partial T}{\partial x} = -\frac{\dot{Q}'''x}{k} = 0 \Rightarrow x=0 \Rightarrow \frac{\partial^2 T}{\partial x^2} = -\frac{\dot{Q}'''}{k} < 0 \quad (T \text{ is a max at } x=0)$$

$$T_{\max} = T_w + \frac{\dot{Q}'''L^2}{2k}$$

$$q'' = q(L) = -k \left. \frac{\partial T}{\partial x} \right|_{x=L} = +k \frac{\dot{Q}'''L}{k} = \dot{Q}'''L$$

$$q''_{\text{out}} = \dot{Q}'''L$$



In general, for heat generation we have a formulation:

$$-\frac{k}{r^n} \frac{\partial}{\partial r} \left(r^n \frac{\partial T}{\partial r} \right) = \dot{Q}''' \quad \begin{cases} n=0 & (\text{slab}) \\ n=1 & (\text{cylinder}) \\ n=2 & (\text{sphere}) \end{cases}$$

This only works if $\dot{Q}''' = \text{constant}$

$$-k \frac{\partial}{\partial r} \left(r^n \frac{\partial T}{\partial r} \right) = \dot{Q}''' r^n$$

$$-k r^n \frac{\partial T}{\partial r} = \dot{Q}''' \int r^n dr = \frac{\dot{Q}'''}{n+1} r^{n+1} + C_1$$

$$q'' = -k \frac{\partial T}{\partial r} = \frac{\dot{Q}''' r}{n+1} + \frac{C_1}{r^n}$$

Choosing $q''(0) = 0 \rightarrow C_1 = 0$ for $n=0$

For $n=1, 2$, $C_1 = 0$ if $|q''| < \infty$ (since $\frac{C_1}{r^1}$ or $\frac{C_1}{r^2}$ at $r \rightarrow 0$ blows up)

$$\frac{\partial T}{\partial r} = -\frac{\dot{Q}''' r}{(n+1)k}$$

$$T = -\frac{\dot{Q}'''}{(n+1)k} \int r dr = C_2 - \frac{\dot{Q}''' r^2}{2(n+1)k}$$

$$\text{Letting } T(a) = T_a \Rightarrow T_a = C_2 - \frac{\dot{Q}''' a^2}{2(n+1)k} \Rightarrow C_2 = T_a + \frac{\dot{Q}''' a^2}{2(n+1)k}$$

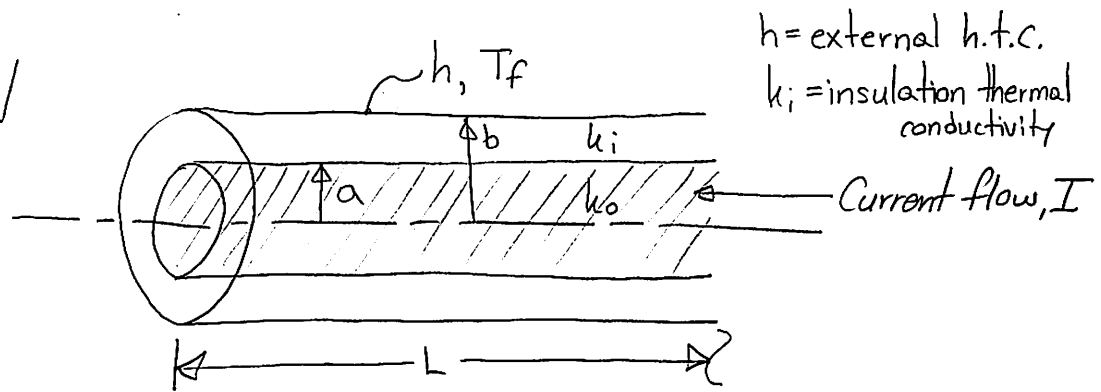
So our generalized result is:

$$T(r) = T_a + \frac{\dot{Q}''' a^2}{2(n+1)k} \left(1 - \frac{r^2}{a^2} \right) \Rightarrow \text{Temperature profile}$$

$$q'' = \frac{\dot{Q}''' r}{n+1} \Rightarrow \text{Heat flux}$$

$$T_{\max} = T(r=0) = T_a + \frac{\dot{Q}''' a^2}{2(n+1)k} \Rightarrow \text{Maximum temperature where } T_a \text{ is the boundary temperature on the outside.}$$

Example #3/

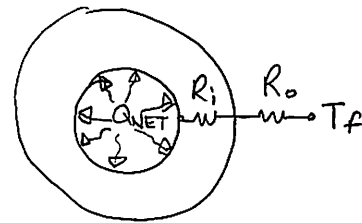


We have the wire electrical resistance, R_e , and the current, I , what is T_{max} ?

$$\text{Energy loss (Joule Heating)} = I^2 R_e$$

$$\dot{Q}''' = \frac{I^2 R_e}{\pi a^2 L}$$

$$Q_{net} = \pi a^2 L \dot{Q}''' = \frac{\Delta T}{R_{TOT}}$$



$$\pi a^2 L \dot{Q}''' = \frac{T_a - T_f}{\frac{\ln(b/a)}{2\pi L k_i} + \frac{1}{2\pi b L h}}$$

$$T_a = T_f + \frac{a^2 \dot{Q}'''}{2} \left[\frac{\ln(b/a)}{k_i} + \frac{1}{bh} \right] \quad (1)$$

We just learned that for a cylinder ($n=1$)

$$T_{max} = T_a + \frac{\dot{Q}''' a^2}{4k} \quad (2)$$

Substitute (1) into (2)

$$T_{max} = T_f + \frac{a^2 \dot{Q}'''}{2} \left[\frac{1}{2k_o} + \frac{\ln(b/a)}{k_i} + \frac{1}{bh} \right]$$